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#include <iostream>
#include <vector>
using namespace std;

vector<vector<int>> matrixReshape(vector<vector<int>>& mat, int r, int c) {
    int m = mat.size();
    int n = mat[0].size();

    // Reshape not possible
    if (m * n != r * c)
        return mat;

    vector<vector<int>> result(r, vector<int>(c));

    for (int i = 0; i < m * n; i++) {
        result[i / c][i % c] = mat[i / n][i % n];
    }

    return result;
}

int main() {
    vector<vector<int>> mat = {
        {1, 2},
        {3, 4}
    };

    int r = 1, c = 4;

    vector<vector<int>> ans = matrixReshape(mat, r, c);

    for (auto &row : ans) {
        for (int x : row)
            cout << x << " ";
        cout << endl;
    }

    return 0;
}

```